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Dear Editorial Board at *Ecology Letters*,

We would like to propose that the manuscript, titled “A synthesis of theoretical understanding and empirical evidence for plant responses to nitrogen and phosphorus supply”, be considered as a *Synthesis* article in *Ecology Letters*.

Carbon (C), nitrogen (N), and phosphorus (P) cycles are tightly coupled in terrestrial ecosystems, and their interactions dictate the magnitude of the terrestrial carbon sink. Although terrestrial biosphere models are beginning to incorporate coupled C-N-P cycles, projections of the land carbon sink still diverge substantially across models, with greater divergence observed among nutrient-coupled models than models that only incorporate the C cycle. This uncertainty stems, in part, from an incomplete understanding of how N and P interact to influence leaf nutrient content, physiology, and biomass allocation, and how these responses scale to regulate ecosystem productivity. Syntheses that resolve these interactions are needed to improve the reliability of terrestrial biosphere models. However, existing syntheses have quantified these interactions for only a few whole-plant traits relevant for terrestrial biosphere modeling, restricting their ability to provide an empirical and theoretical basis for improving model performance and reliability. To address this gap, we compiled 5,680 observations from 90 full-factorial N-by-P supply experiments worldwide. We used this dataset to quantify the individual and interactive effects of N and P supply on 19 leaf and whole-plant variables that are directly relevant for assessing model performance and reliability and refining ecophysiological theory.

This synthesis establishes several novel principles for understanding C-N-P interactions. First, N and P supply increased leaf nutrient content and biomass production more strongly than it did photosynthetic variables, highlighting the need for models to relax assumptions about constant leaf nutrient-photosynthesis relationships. Second, interaction effect sizes indicated largely additive leaf-level responses to combined N and P supply, highlighting the idea that N and P play distinct and independent roles on leaf nutrient and photosynthetic processes. Third, our synthesis shows strong climate controls on leaf nutrient responses to

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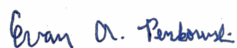
nutrient enrichment, supporting the idea that environments with greater photosynthetic demand for soil resources (i.e., cold, dry, and bright environments) experience stronger responses to nutrient supply. These patterns highlight the utility of including climate-based frameworks for predicting photosynthetic investment in models. Fourth, our results show that nutrient acquisition strategy is more informative than photosynthetic pathway when modeling plant responses to nutrient supply, supporting the inclusion and further development of acquisition strategy-explicit frameworks in models. Overall, this synthesis provides strong empirical support for an emerging ecophysiological-grounded theoretical framework for C-N-P cycling, which in turn provides a basis for improving the representation of nutrient cycles in terrestrial biosphere models.

This work is a direct follow-up to a recently published *Tansley Review* in *New Phytologist* (Stocker et al., 2025) that was just recognized as being one of the journal's top 10 most viewed articles of 2025. The same core set of authors contributed substantially to this proposed *Synthesis*. Given this and the novel insights this synthesis provides, we firmly believe that this synthesis will be of broad interest to plant ecologists, ecosystem ecologists, and model developers.

Thank you for considering this proposal. We would be happy to submit this manuscript as a *Letter* or *Perspective* if the editorial team deems these article types more suitable and are interested in submitting this manuscript for the *Ecological and Evolutionary Responses to Multiple Stressors* special feature regardless of article type.

Please contact me using the e-mail listed above with any questions or concerns.

Sincerely,



Evan A. Perkowski, Ph.D.

On behalf of co-authors Alissar Cheaib, Monika R. Kelley, Jan A. Lankhorst, Astrid Odé, Daniil J.P. Scheifes, Huiying Xu, Daisy Cárate-Tandalla, Hugo J. de Boer, Ning Dong, Benjamin D. Stocker, Karin T. Rebel, I. Colin Prentice, Sandy P. Harrison, and Nicholas G. Smith

References

Stocker, B.D., Dong, N., Perkowski, E.A., Schneider, P.D., Xu, H., de Boer, H.J., et al. (2025). Empirical evidence and theoretical understanding of ecosystem carbon and nitrogen cycle interactions. *New Phytologist*, 245, 49–68.

